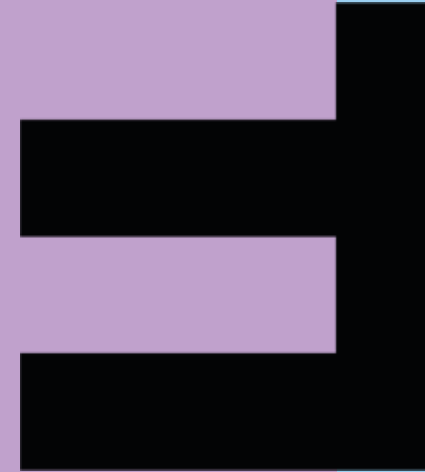


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of Applied Sciences



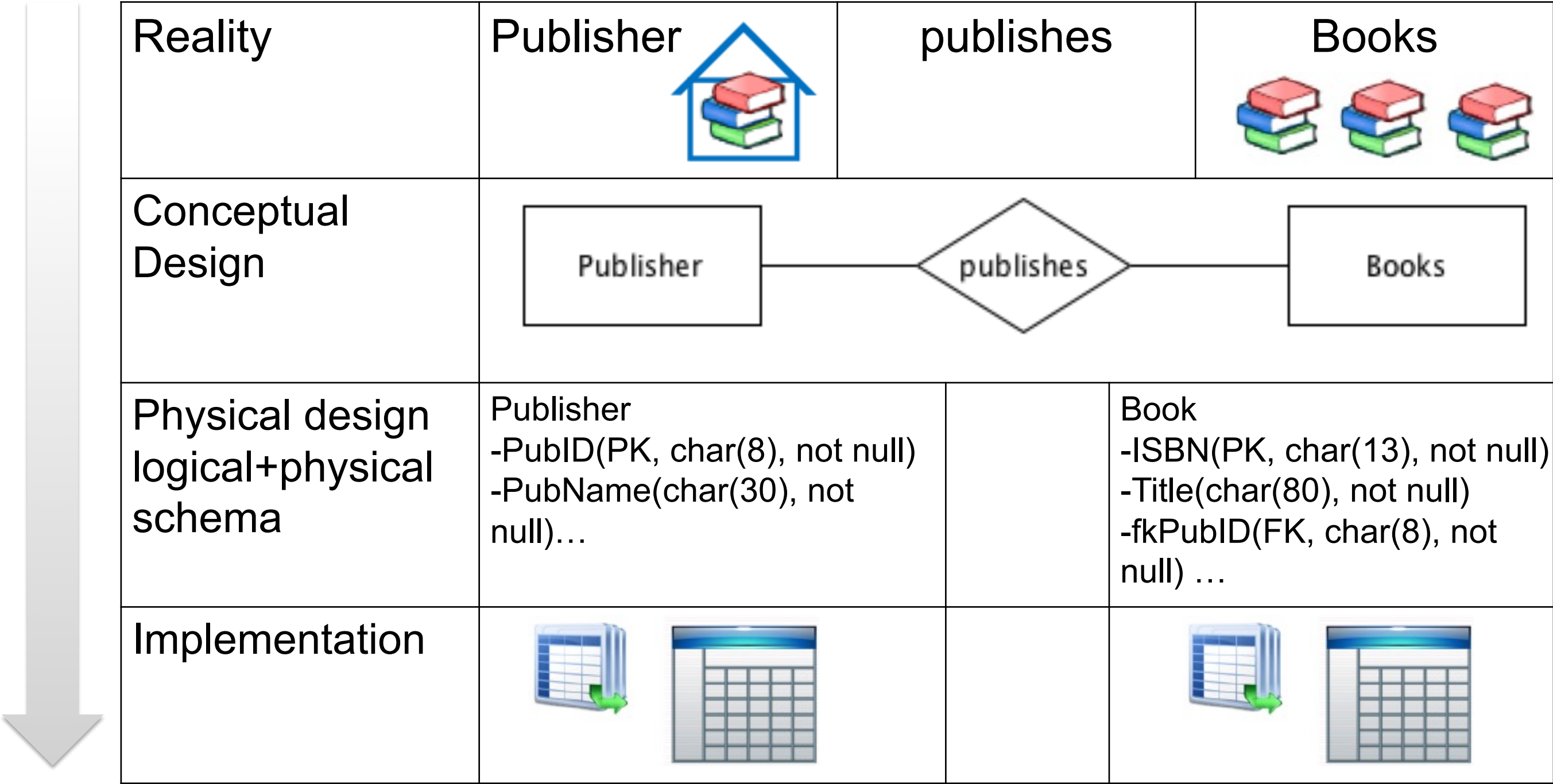
Technical Information Systems

3. Normalization, Database Design

3. Normalization, Database Design

1. ERM Modelling Process

The data modelling process



source: Dietmar Grabher – WIN Lab

Data Modelling – Entity-Relationship-Modelling

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- Determining the structure in 3 steps



1. First examine your information to find the components of the UoD, that are

✧ **distinct** and

✧ **specific**

Objects of our reality or our conception.

These objects are called **Entities**.



Data Modelling – Entity-Relationship-Modelling

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- Determining the structure in 3 steps



2. Now we examine the **relationships** between the entities. How are they connected?
Do they interact? Does one not exist without the other?

In a second run we examine the possible number of entities that can be connected.

Are there one or many entities (→ cardinality)?

Is this connection a must (→ optionality)



Data Modelling – Entity-Relationship-Modelling

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- Determining the structure in 3 steps



3. Lastly we analyse the **properties** of our entities.

What kind of characteristics can be identified?

What is **relevant** for our UoD?

These characteristics translate in to attributes of the entities.

We identify

- ✧ The **kind** of attribute
- ✧ The **domain** (range of values)

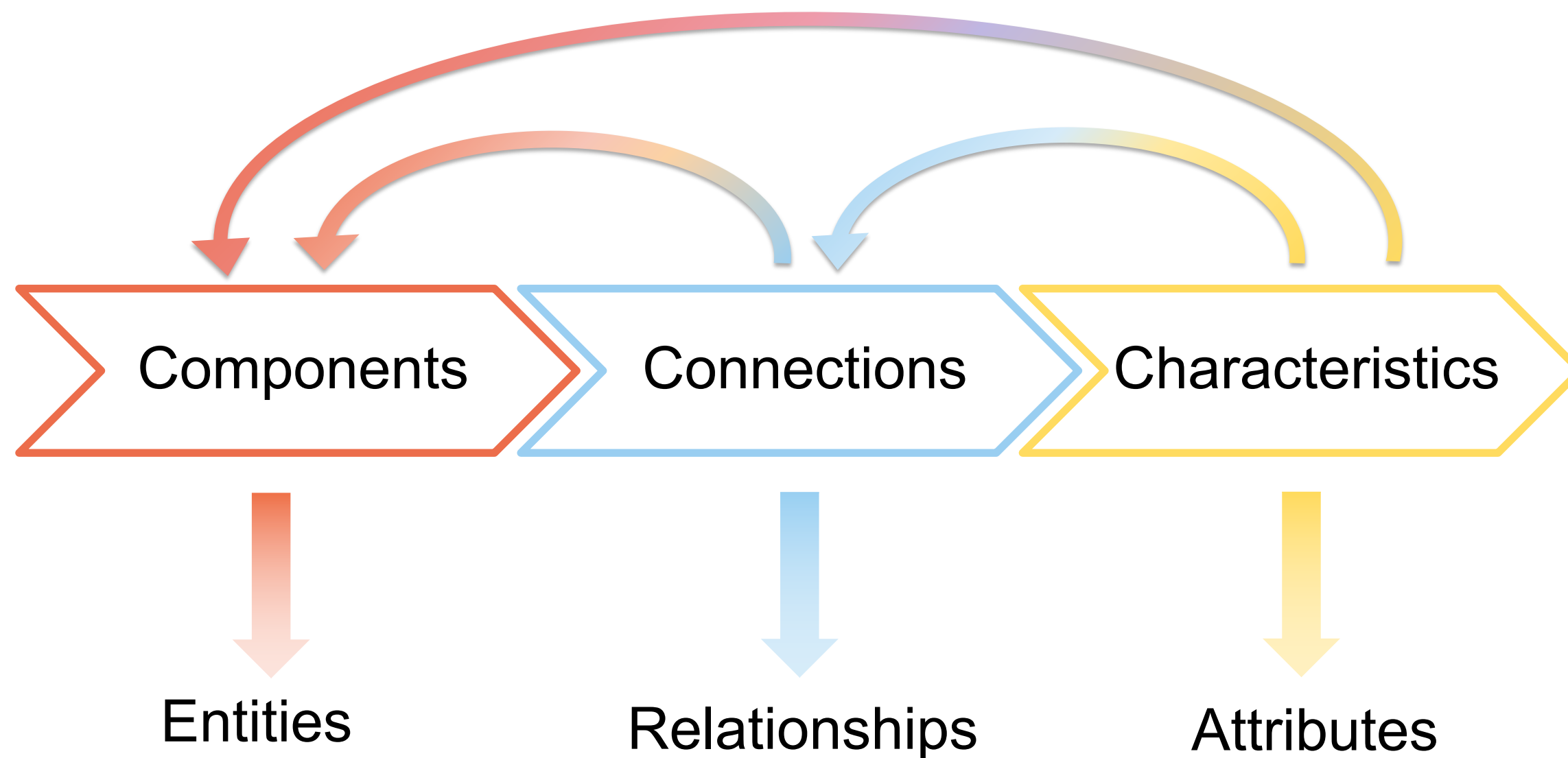


Data Modelling – Entity-Relationship-Modelling

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- Determining the structure in 3 steps



Data Modelling – From Conceptual Design to Physical Design

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1. Each entity becomes a table
2. Each attribute of an entity becomes an attribute in the respective table (Data formats as demanded through the requirements).
Complex attributes must be split into scalar attributes (or even model extensions → new entities).
3. If there is no good natural primary key, a surrogate key is added.
4. A 1:1 relationship leads to a foreign key in the subordinate table to the main table. Often, this foreign key will also be the primary key of the table.
5. A 1:n relationship leads to a foreign key in the table belonging to the n-entity
6. An n:m relationship leads to an interim table with foreign keys to the two entities. Primary key are both foreign keys or a surrogate key. Attributes belonging to the n:m relationship are created in the interim table (reification)

**Time for some
exercises on
modeling**

Unified Modeling Language (UML)

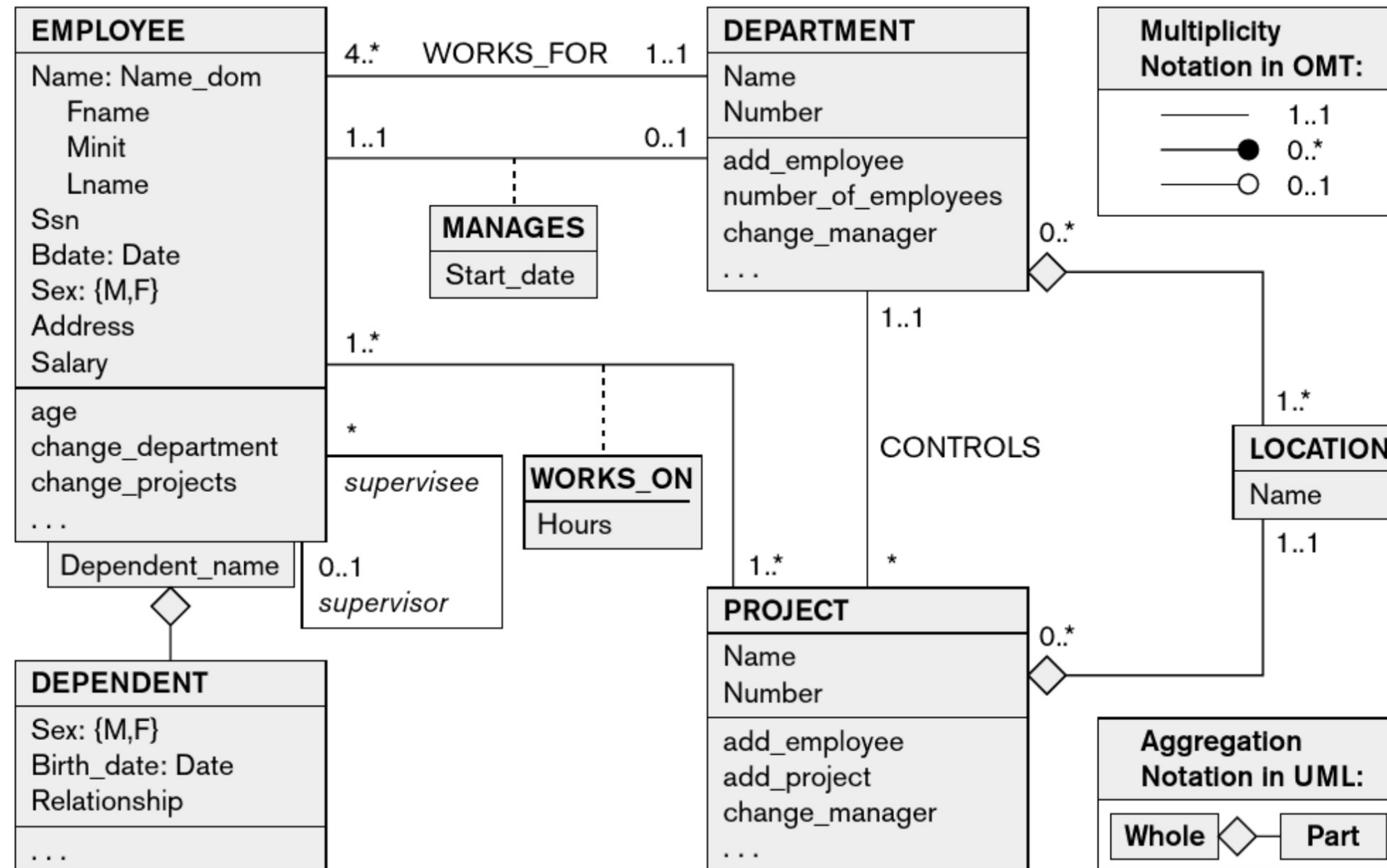
- There is a significant demand to cover the entire spectrum from requirements analysis, modelling, design and implementation to the deployment of database-based applications with visual tools.
- The Unified Modeling Language (UML) developed by the Object Management Group (OMG) is currently the best accepted standard for this purpose.
- UML provides a standard visual notation and associated language for all the above mentioned tasks.
- UML can be used by developers, data modellers, database developers, etc. to create detailed specifications.
- UML can also be used to implement the entire system environment with users, software, communication and hardware.
- This versatility is also its downside. UML is highly complex and difficult for laypeople to understand.

[EINa17:90]

Unified Modeling Language (UML)

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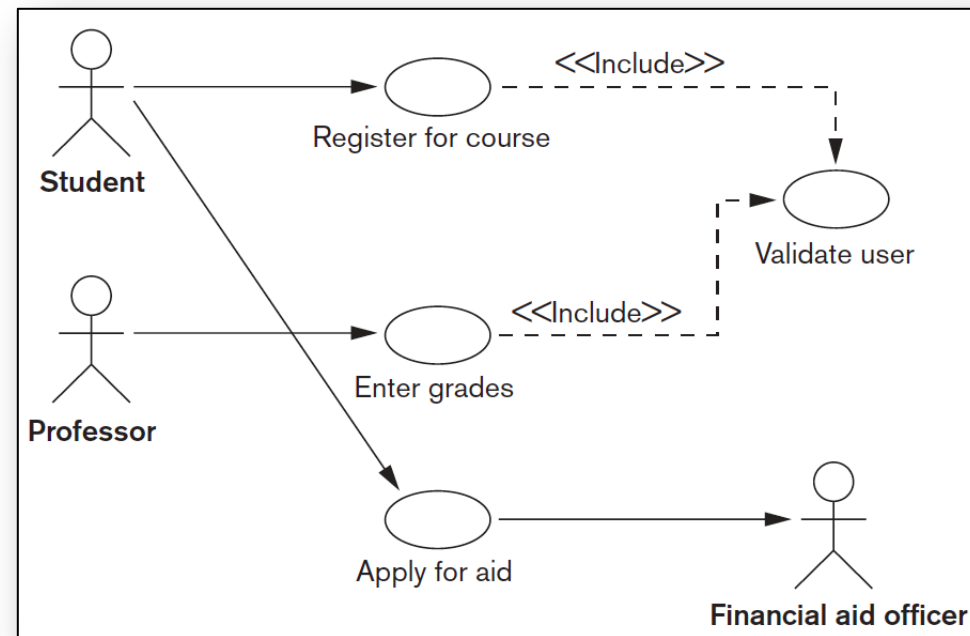
Seite 12



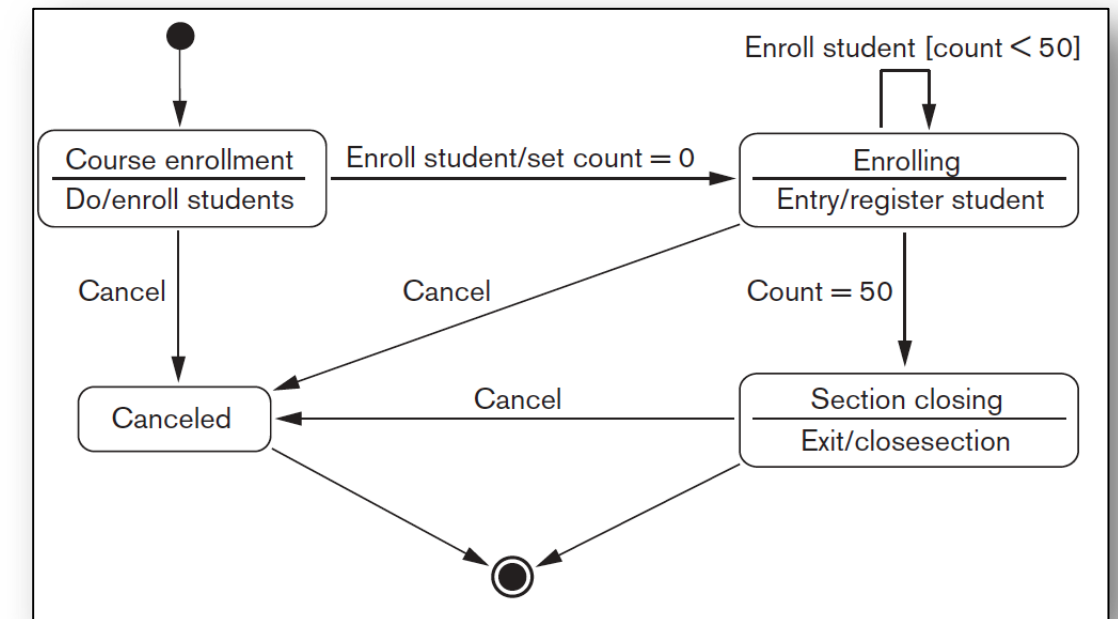
Semantic modelling in UML → Class diagram

[EINa17:115]

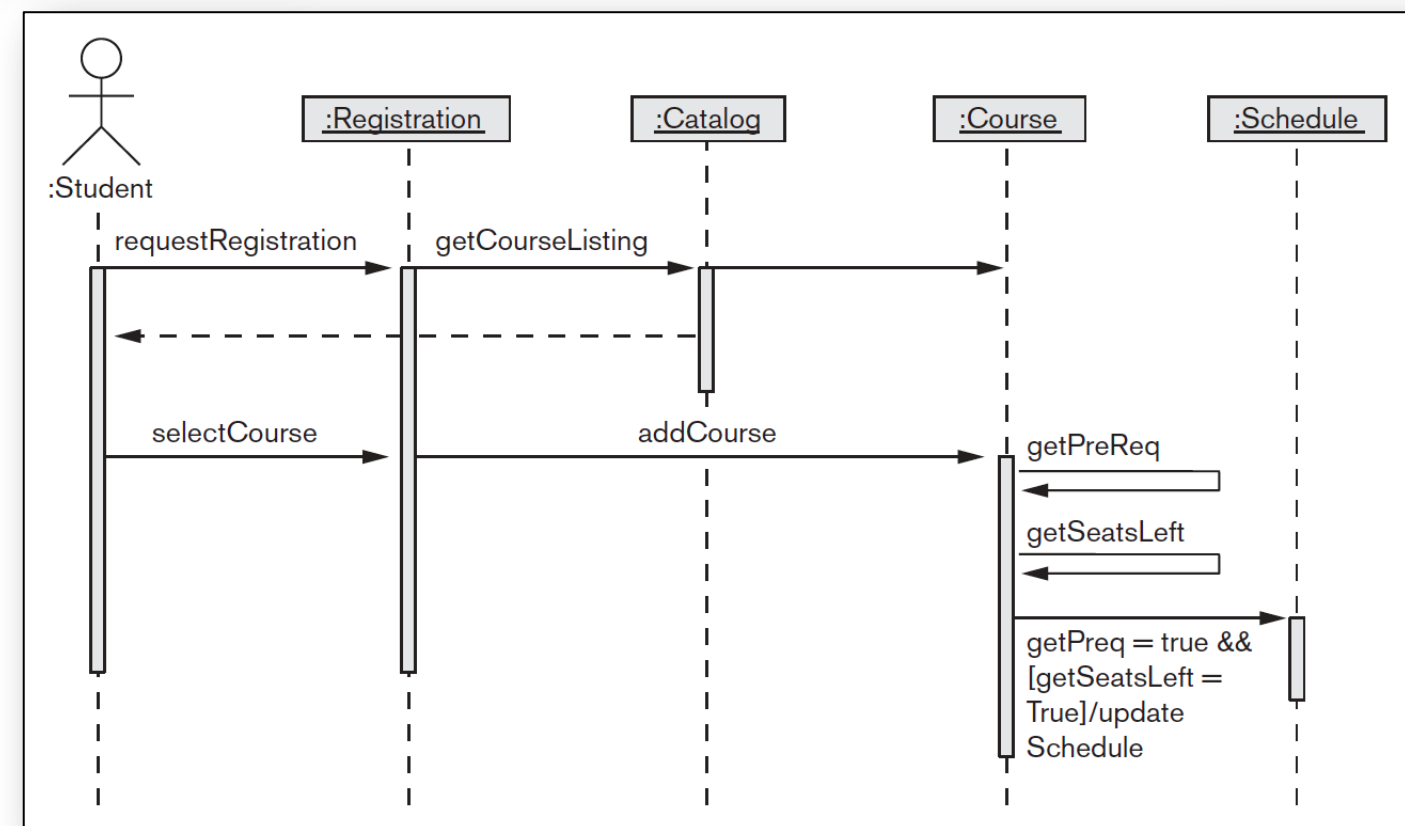
Unified Modeling Language (UML)



use case diagram



statechart diagram



sequence diagram

Database design - comparison of the basic terms

Table	Relationale Datenbank	Entity-Relationship-Modell (ERM)	Unified Modeling Language (UML)
Value range (Domain)	Value range (Domain)	Value range (Domain)	Value range (Domain)
Header	Relation type/ Relation format/ Relation schema	Entity type	Class
Column header	Attribute	Attribute	Attribute
--	Foreign key connection	Relationship	Association
Row	Tuple	Entity	Object, Instance
Cell	Scalar	Attribute value	Attribute value

Datenbankentwurf- Gegenüberstellung der Grundbegriffe

Tabelle	Relationale Datenbank	Entity-Relationship-Modell (ERM)	Unified Modeling Language (UML)
Wertebereich (Domäne, Domain)	Wertebereich (Domäne, Domain)	Wertebereich (Domäne, Domain)	Wertebereich (Domäne, Domain)
Kopfzeile	Relationstyp/ Relationsformat/ Relationenschema	Entitätstyp	Klasse
Spaltenüberschrift	Attribut	Attribut	Attribut
--	Fremdschlüssel- beziehung	Beziehung (Relationship)	Assoziation
Zeile	Tupel	Entität	Objekt, Instanz
Zelle	Skalar, Attributwert	Attributwert	Attributwert

Quelle: [https://de.wikipedia.org/wiki/Entit%C3%A4t_\(Informatik\)](https://de.wikipedia.org/wiki/Entit%C3%A4t_(Informatik))

3. Normalization, Database Design

1. ERM Modelling Process

2. Normalization

Insertion Anomalies

- Insertion of data or an entity can only be done, if data to an other entity is added
- Dependent data can't be inserted, if there is no superordinate entity

Deletion Anomalies

- If superordinate data is deleted, the dependent information vanishes with it

Modification Anomalies

- If redundant data changes there is a big chance, that the modification is not made everywhere and inconsistencies occur
- Same is true when errors are made during data entry

- A good data model avoids repetition of data
- Still, a beginner's data model or some legacy data you might inherit, can include redundant data that needs to be eliminated
- The structured, semi-formal method to efficiently create a good data model is **Normalization**

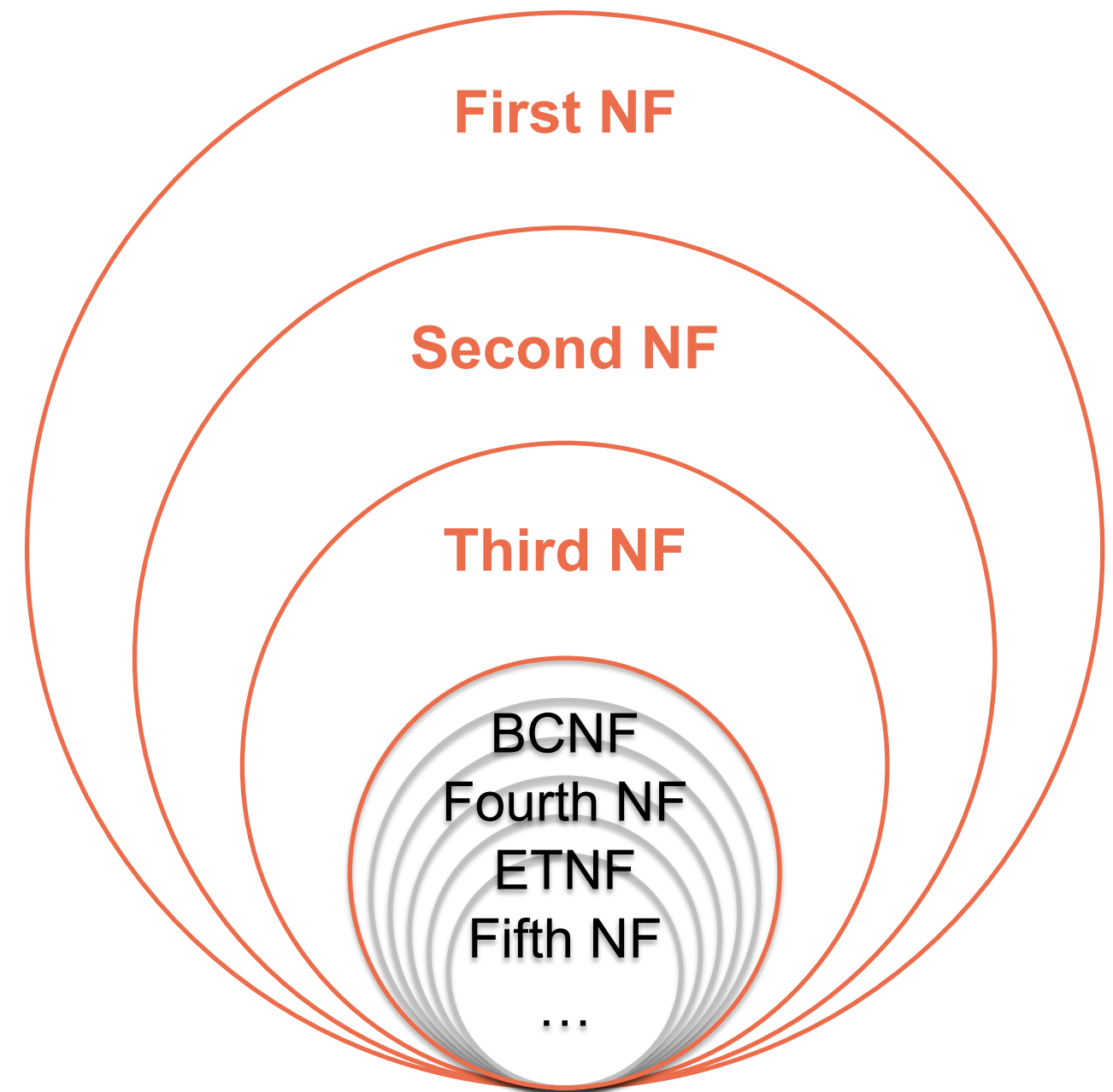
Bill_Nr	Client_Name	Street	Zip	City	Tax_Number	Billing_Date	Price
1	John Doe	Anystreet 12	1234	Anytown	123 456 789	01.08.2016	€575,00
2	Sue Someone	Someplace 13	5678	Somewhere	761 650 745	21.08.2016	€979,00
3	Peter Pan	Tree 14	9876	Neverland	287 910 543	13.08.2016	€34,00
4	P. Sherman	42 Wallaby Way	2222	Sydney	372 442 250	15.08.2016	€631,00
5	Vernon Dursley	Privet Drive 4	3333	Little Whinging	851 064 542	17.08.2016	€970,00
6	Sue Someone	Someplace 13	5678	Somewhere	761 650 745	19.08.2016	€533,00
7	Vernon Dursley	Privet Drive 4	3333	Little Whinging	851 064 542	01.08.2016	€756,00
8	Sue Someone	Someplace 13	5678	Somewhere	761 650 745	23.08.2016	€812,00
9	Vernon Dursley	Privet Drive 4	3333	Little Whinging	851 064 542	22.08.2016	€721,00
11	Vernon Dursley	Privet Drive 4	3333	Little Whinging	851 064 542	15.08.2016	€655,00

Normalization

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- Normalization generates so called normal forms
- A good data model has at least second normal form – better third normal form
- Normal forms do not eliminate all redundancies. They reduce them as much as feasible
- It is part of the design process to decide the depth of the normalization



Normalization – Dependencies

Functional Dependency

- A functional dependency is a constraint between two sets of attributes from the database.
- We regard a set of n attributes $A_1, A_2, \dots, A_n$ in a so-called universal relation $R = \{A_1, A_2, \dots, A_n\}$.
- Definition:
A **functional dependency**, denoted by $X \rightarrow Y$, between two sets of attributes X and Y that are subsets of R specifies a *constraint* on the possible tuples that can form a relation state r of R . The constraint is that, for any two tuples t_1 and t_2 in r that have $t_1[X] = t_2[X]$, they must also have $t_1[Y] = t_2[Y]$.

Normalization – Dependencies

Functional Dependency

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- **Functional dependency:**

In other words: The values of the Y component of a tuple in r depend on, or are determined by, the values of the X component;

$$X \rightarrow Y$$

<u>ArticleNr</u>	Product	Brand	Price
1	Teddy	Steiff	14,99 €
2	Dog	Steiff	7,99 €
3	Cat	Russ Berry	12,00 €
4	Mouse	Russ Berry	4,99 €
5	Teddy	Russ Berry	9,99 €

X

Y

Normalization – Dependencies

Full Functional Dependency

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■ Full functional dependency

"A set of attributes Y is **fully dependent** from a set of attributes X , if Y is dependent from X but not already from a part of X

Full functional dependency

<u>ClientNr</u>	<u>CarType</u>	PurchDate	ClientName
11	1	17.2.18	Meier
12	2	17.2.18	Muller
13	2	18.2.18	Schmidt
11	3	19.2.18	Meier
14	4	20.2.18	Fitz

Partial functional dependency

Normalization – Dependencies

Transitive Dependency

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■ Transitive Dependency:

A set of attributes Z is transitive dependent from a set of attributes X , if Y is functionally dependent from X and Z is functionally dependent from Y

<u>PlayerNr</u>	PlayerName	ClubName	ClubTown
M1	Alaba	FC Bayern München	München
M2	Baumgartlinger	FC Bayer Leverkusen	Leverkusen
M3	Goiginger	LASK	Linz
M4	Grillitsch	TSG 1899 Hoffenheim	Hoffenheim
M5	Ilsanker	RB Leipzig	Leipzig



First normal form

- A table has First Normal Form when
 - All attribute values are atomic (or indivisible), simple, and unstructured
 - There are not repetitive attributes (e.g. phone1, phone2)
 - There is a Primary Key

LECTURE_ID	DTSTART	DTSTART_time	DTEND	DTEND_time	LOCATION	Category	Module	Lecturer	Group
20161010T090000	10.10.16	09:00:00	10.10.16	12:15:00	U426 pc-Pool	SE	Object Oriented Programming	Geiger Franz Schwarz Thomas	EDB5
20161011T095000	11.10.16	09:50:00	11.10.16	12:15:00	U130 eLab	SE	PLC Programming	Amann Robert	EDB5 EDB5_In
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U304, U305	SE	Energy Transmission	Feustel Felix Nenning Andreas	EDB5 EDB5_In
20161017T140000	17.10.16	14:00:00	17.10.16	17:15:00	U427 3D/CAD Lab/pc-Pool	SE	Control Engineering	Feurstein Rudi	EDB5 EDB5_In
20161017T090000	17.10.16	09:00:00	17.10.16	12:15:00	U426 pc-Pool	SE	Object Oriented Programming	Geiger Franz Schwarz Thomas	EDB5
20161018T095000	18.10.16	09:50:00	18.10.16	12:15:00	U130 eLab	SE	PLC Programming	Amann Robert	EDB5 EDB5_In
20161018T140000	18.10.16	14:00:00	18.10.16	17:15:00	U304	SE	Energy Transmission	Feustel Felix Nenning Andreas	EDB5 EDB5_In
20161019T095000	19.10.16	09:50:00	19.10.16	13:05:00	U429	SE	Energy Systems	Nenning Andreas	EDB5 EDB5_In
20161019T140000	19.10.16	14:00:00	19.10.16	17:15:00	U406	SE	Drive Engineering	Schwarz Thomas	EDB5 EDB5_In
20161020T095000	20.10.16	09:50:00	20.10.16	13:05:00	U131 Embedded Systems Lab	SE	Hardware Description Languages	Mitterbacher André	EDB5 EDB5_In
20161020T173000	20.10.16	17:30:00	20.10.16	18:30:00	W206 (mit Gefaelle)	VER	Technikforum	Pilsan Horatiu	EDB1 EDB3 EDB5 ETW1 ETW3 ITB1 ITB3 ITB5
20161021T140000	21.10.16	14:00:00	21.10.16	17:15:00	U304	SE	Electrical Power Systems	Rocks Alexander	EDB5 EDB5_In

First normal form

- A table has First Normal Form when
 - All attribute values are atomic (or indivisible), simple, and unstructured
 - There are not repetitive attributes (e.g. phone1, phone2)
 - There is a Primary Key

LECTURE_ID	DTSTART	DTSTART_time	DTEND	DTEND_time	LOCATION	LOC-info	Category	Module	Lecturer	Group
20161010T090000	10.10.16	09:00:00	10.10.16	12:15:00	U426	pc-Pool	SE	Object Oriented Programming	Geiger Franz	EDB5
20161010T090000	10.10.16	09:00:00	10.10.16	12:15:00	U426	pc-Pool	SE	Object Oriented Programming	Schwarz Thomas	EDB5
20161011T095000	11.10.16	09:50:00	11.10.16	12:15:00	U130	eLab	SE	PLC Programming	Amann Robert	EDB5
20161011T095000	11.10.16	09:50:00	11.10.16	12:15:00	U130	eLab	SE	PLC Programming	Amann Robert	EDB5_In
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U304		SE	Energy Transmission	Feustel Felix	EDB5
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U304		SE	Energy Transmission	Nenning Andreas	EDB5
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U304		SE	Energy Transmission	Feustel Felix	EDB5_In
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U304		SE	Energy Transmission	Nenning Andreas	EDB5_In
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U305		SE	Energy Transmission	Feustel Felix	EDB5
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U305		SE	Energy Transmission	Nenning Andreas	EDB5
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U305		SE	Energy Transmission	Feustel Felix	EDB5_In
20161011T140000	11.10.16	14:00:00	11.10.16	17:15:00	U305		SE	Energy Transmission	Nenning Andreas	EDB5_In

All the red attributes are needed to form the Primary Key

First normal form – second example

Not normalized

CD_ID	Album	Founding_Year	Titles
4711	Anastacia – Not That Kind	2000	{1. Not That Kind, 2. I’m Outta Love, 3. Cowboys & Kisses}
4712	Pink Floyd – Wish You Were Here	1975	{1. Shine On You Crazy Diamond}
4713	Anastacia – Freak of Nature	2001	{1. Paid my Dues}

First NF

CD_ID	Album_Title	Artist	Founding_Year	Track	Title
4711	Not That Kind	Anastacia	2000	1	Not That Kind
4711	Not That Kind	Anastacia	2000	2	I’m Outta Love
4711	Not That Kind	Anastacia	2000	3	Cowboys & Kisses
4712	Wish You Were Here	Pink Floyd	1975	1	Shine On You Crazy Diamond
4713	Freak of Nature	Anastacia	2001	1	Paid my Dues

Datasource: [https://de.wikipedia.org/wiki/Normalisierung_\(Datenbank\)#Zweite_Normalform_.282NF.29](https://de.wikipedia.org/wiki/Normalisierung_(Datenbank)#Zweite_Normalform_.282NF.29)

Normalization – Second Normal Form

- A table has **Second Normal Form** when
 - the table is in **first normal form** and
 - all attributes are **fully dependent** from the whole primary key
- First NF

CD_ID	Album_Title	Artist	Founding_Year	Track	Title
4711	Not That Kind	Anastacia	1999	1	Not That Kind
4711	Not That Kind	Anastacia	1999	2	I'm Outta Love
4711	Not That Kind	Anastacia	1999	3	Cowboys & Kisses
4712	Wish You Were Here	Pink Floyd	1965	1	Shine On You Crazy Diamond
4713	Freak of Nature	Anastacia	1999	1	Paid my Dues

- Second NF

CD_ID	Album_Title	Artist	Founding_Year
4711	Not That Kind	Anastacia	1999
4712	Wish You Were Here	Pink Floyd	1965
4713	Freak of Nature	Anastacia	1999

CD_ID	Track	Title
4711	1	Not That Kind
4711	2	I'm Outta Love
4711	3	Cowboys & Kisses
4712	1	Shine On You Crazy Diamond
4713	1	Paid my Dues

Table



Lossless
Decomposition



Projections

Normalization – Third Normal Form

- A table has **Third Normal Form** when
 - the table is in **second normal** form and
 - no attribute of the table is **transitively dependent** the primary key.

- Second NF



CD_ID	Album Title	Artist	Founding Year
4711	Not That Kind	Anastacia	1999
4712	Wish You Were Here	Pink Floyd	1965
4713	Freak of Nature	Anastacia	1999

- Third NF



Artist ID	Artist	Founding Year
311	Anastacia	1999
312	Pink Floyd	1965

CD_ID	Album Title	Artist ID
4711	Not That Kind	311
4712	Wish You Were Here	312
4713	Freak of Nature	311

CD_ID	Track	Title
4711	1	Not That Kind
4711	2	I'm Outta Love
4711	3	Cowboys & Kisses
4712	1	Shine On You Crazy Diamond
4713	1	Paid my Dues

[EINa16:513]

Additional Example – Pre-normalisation

<u>PlayerId</u>	Player	Hometown	ClubId	ClubName	Observer Id	ObserverN ame	Grade
101	Häßler	München	9	München 1860	50/51	Merkel/ Breitner	2/5
102	Kirsten	Neuss	10	Bayer Leverkuse n	52	Ribbeck	3
103	Bobic	Bochum	11	Borussia Dortmund	50/51/52	Merkel/Breit- ner/Ribbeck	3/4/3
104	Kohler	Herne	11	Borussia Dortmund	50/52	Merkel/ Ribbeck	2/3

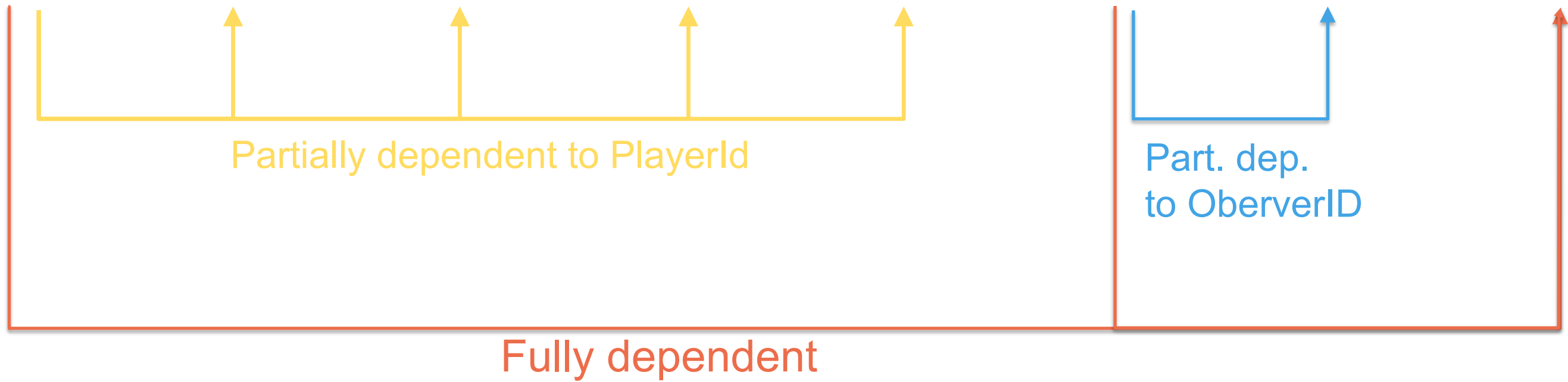
Example: [Gada19:39ff]

Additional Example – First Normal Form

PlayerId	Player	Hometown	ClubId	ClubName	ObserverId	ObserverName	Grade
101	Häßler	München	9	München 1860	50	Merkel	2
101	Häßler	München	9	München 1860	51	Breitner	5
102	Kirsten	Neuss	10	Bayer Leverkusen	52	Ribbeck	3
103	Bobic	Bochum	11	Borussia Dortmund	50	Merkel	3
103	Bobic	Bochum	11	Borussia Dortmund	51	Breitner	4
103	Bobic	Bochum	11	Borussia Dortmund	52	Ribbeck	3
104	Kohler	Herne	11	Borussia Dortmund	50	Merkel	2
104	Kohler	Herne	11	Borussia Dortmund	52	Ribbeck	3

Additional Example – Towards Second Normal Form

<u>PlayerId</u>	Player	Hometown	ClubId	ClubName	ObserverId	ObserverName	Grade
101	Häßler	München	9	München 1860	50	Merkel	2
101	Häßler	München	9	München 1860	51	Breitner	5
102	Kirsten	Neuss	10	Bayer Leverkusen	52	Ribbeck	3



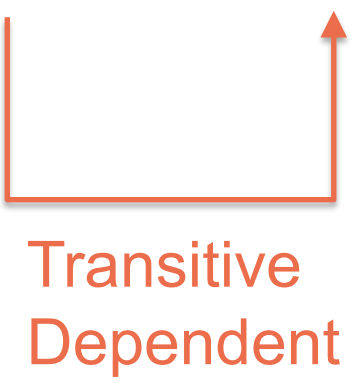
Additional Example – Second Normal Form

Player

<u>PlayerId</u>	Player	Hometown	ClubId	ClubName
101	Häßler	München	9	München 1860
102	Kirsten	Neuss	10	Bayer Leverkusen
103	Bobic	Bochum	11	Borussia Dortmund
104	Kohler	Herne	11	Borussia Dortmund

Observer

<u>ObserverId</u>	ObserverName
50	Merkel
51	Breitner
52	Ribbeck



PlayerObserver

<u>PlayerId</u>	<u>ObsId</u>	Grade
101	50	2
101	51	5
102	52	3
103	50	3
103	51	4
103	52	3
104	50	2
104	52	3

Additional Example – Third Normal Form

Player

<u>P.-Id</u>	Player	H.-town	ClubId
101	Häßler	München	9
102	Kirsten	Neuss	10
103	Bobic	Bochum	11
104	Kohler	Herne	11

Observer

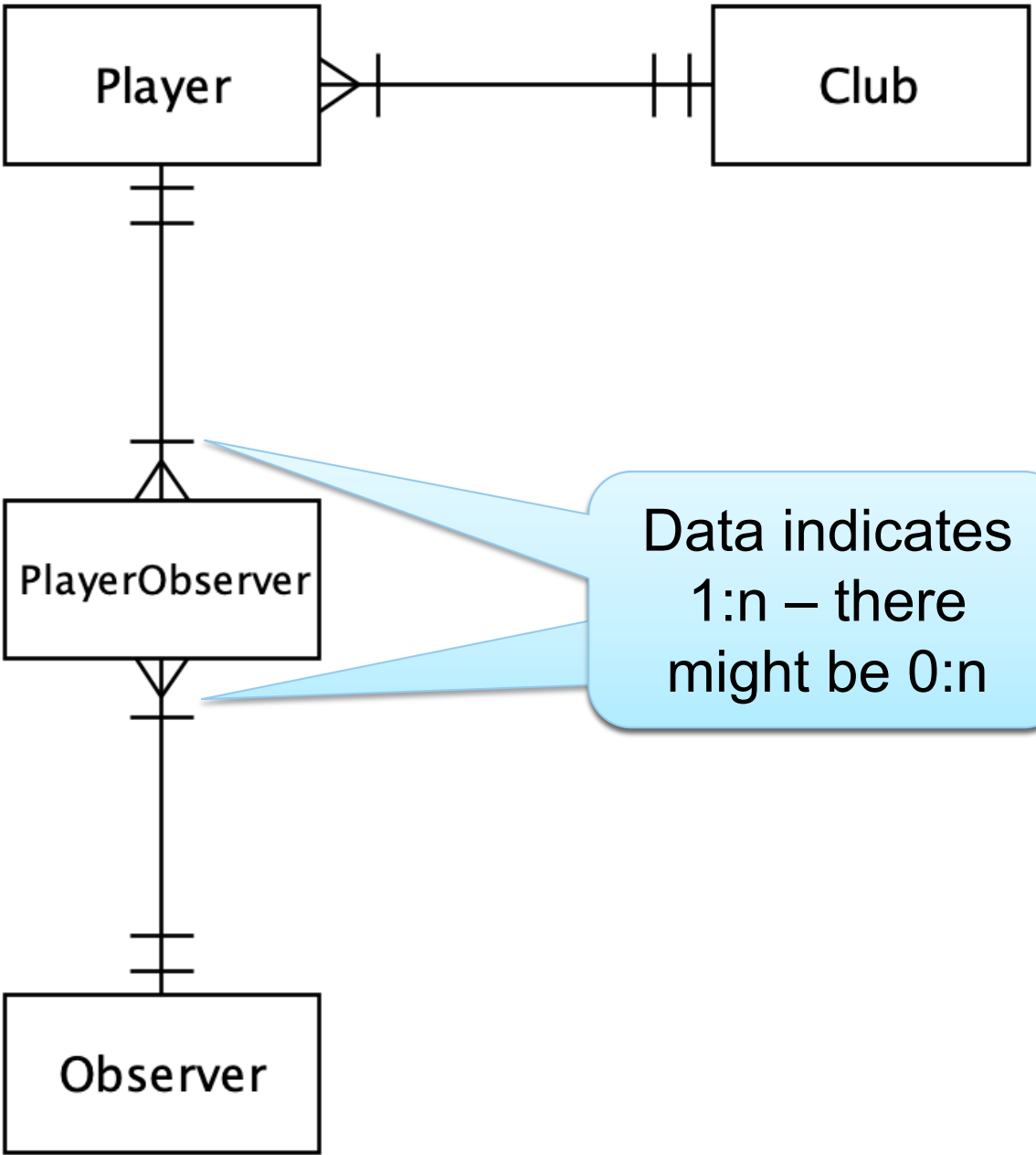
<u>O.-Id</u>	ObserverName
50	Merkel
51	Breitner
52	Ribbeck

PlayerObserver

<u>PlayerId</u>	<u>ObsId</u>	Grade
101	50	2
101	51	5
102	52	3
103	50	3
103	51	4
103	52	3
104	50	2

Club

<u>ClubId</u>	ClubName
9	München1860
10	Bayer Leverkusen
11	Borussia Dortmund
11	Borussia Dortmund



**Time for more
exercises**

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